

A Packing of 17 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.832993. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.823105, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.832993 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 17$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.823105, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 17/16 = 2.833333$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N17.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 66 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
944331305847816078184053810523473169925604273230942657736554520745155497
506755826736768012231950882663138575000453397516763973452895588811388385
959279892048015370480259087800662115454749373957
```

denominator:

```
33333333333249784299232111407714619686402613898186761660255982748649841
262465915175292007964718597733564623435487238291620011720625015854228562
594578487217764254802867354938249283666473879424
```

B Coordinates

17 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N17.txt`.

```
79469527432 94011962216 259955481113 386763734143 -43036780190 -99630245329 530095890795 186652025248-7774673997 67605866989 34257414762 486606497425
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